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10 February 1984
SUPERSEDING
MIL-A-52424A(ME)
29 May 1967

MILITARY SPECIFICATION

AIR CONDITIONERS: WALL- OR BASE-MOUNTED, SELF-CONTAINED, MULTIPACKAGE, STANDARD WEIGHT, AIR-COOLED, 6,000-BTU/HR

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained, multi-package air conditioners. The air conditioners control temperatures and humidities and are suitable for use in cold or hot and humid conditions.

1.2 Classification. The air conditioners shall be of the following classes, as specified (see 6.2):

Class 1 - 115-volt, single-phase, 2-wire, 50/60-hertz.
Class 2 - 208-volt, 3-phase, 4-wire, 400-hertz.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4120

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SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification, Instruction, and Blank Plate.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810	- Environmental Test Methods for Aerospace and Ground Equipment.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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DRAWINGS

ME

DL13211E4726	- Air Conditioner, Multipackage, 6,000 Btu/Hr, Model CE 6A-60.
DL13211E4600	- Air Conditioner, Multipackage, 6,000 Btu/Hr, Model CE 6A-400.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AIR CONDITIONING AND REFRIGERATION INSTITUTE

Standard 515-60 - Standard for Sealed Refigerant Compressors and Condensing Units, 20 HP and Smaller.

(Application for copies should be addressed to the Air Conditioning and Refrigerant Institute, 1815 North Fort Myer Drive, Arlington, VA 22209.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard 16-69 - Method of Testing for Rating Room Air Conditioners.

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 345, East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechancial Engineers, 345 East 47th Street, New York, NY 10017.)

SOUTH COAST QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents

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(Application for copies should be addressed to the South Coast Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The air conditioners shall be in accordance with the applicable data lists and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissable without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the air conditioners. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2) the contractor shall furnish two air conditioners within the time frame specified in the contract or purchase order to prove prior to starting production, that his production methods will produce air conditioners that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the first article during production shall be subject to the approval of the contracting officer. Approval of the first article by the activity concerned shall not relieve the contractor of his obligation to furnish air conditioners conforming to this specification.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more air conditioners (see 6.2) for testing as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.4.1 Material deterioration and control. Material not specified in fabrication of the air conditioner shall be compatible to the specified materials and shall be inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

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3.4.1.1 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make this information available, upon request, to the contracting officer or designated representative.

3.5 Performance characteristics.

3.5.1 Capacity. When operating at sea level, the total net cooling capacity shall be not less than 6,000 Btu/hr for class 2 air conditioners and 5,800 Btu/Hr for class 1 air conditioners under each of the sets of climatic conditions shown in table I. Class 1 air conditioners shall be capacity rated when operated on 115-volt, single-phase, 60-hertz input. Class 2 air conditioners shall be capacity rated when operated on 208-volt, 3-phase, 400-hertz input.

TABLE I. Climatic conditions for capacity rating.

Condition	Ambient Temperature °F	Evaporator Air Inlet Temperature °F	
	Dry Bulb	Dry Bulb	Wet Bulb
I	125	90	75
II	95	80	67

3.5.2 Operating temperatures. The air conditioner operating on 50-hertz input for class 1 and 400-hertz input for class 2 shall function as follows:

a. Start, operate, and cycle manually on cooling cycle at each set of climatic conditions specified in table I.
b. Start, operate, and cycle manually on cooling cycle at +125° F ambient temperature with air at +125° F temperature entering the evaporator.
c. Operate on cooling cycle without forming frost or ice on the evaporator coil at an ambient temperature of +60° F with air at +75° F dry bulb and +63° F wet bulb entering the evaporator.
d. Operate on heating cycle in ambient temperature as low as -65° F and as high as +80° F.

3.5.3 Pressures. When the air conditioner is operating as specified in 3.5.2, the condensing pressure shall be not more than 270 pounds per square inch gage (psig) and the suction pressure shall be not less than 40 psig when the evaporator unit and condensing unit are joined by not more than 4 feet of refrigerant lines.

3.5.4 Evaporator airflow. When the air conditioner is operating at the conditions specified in 3.5.1, but with the refrigeration system inoperative, the evaporator airflow to the space being cooled shall be within the range of 200 to 240 standard cubic feet per minute (scfm) free air delivery.

3.5.5 Cooling ratio. When operating at the conditions specified in 3.5.1, the air conditoner shall have a sensible cooling capacity between 70 and 90 percent of the total net cooling capacity.

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3.5.6 Power consumption. When the air conditioner is operating at the conditions specified in 3.5.1, the total power consumption for class 1 shall not exceed 1.8 kilowatts per hour (kwh) and for class 2 shall not exceed 2.0 kwh.

3.5.7 Heating. With all heaters operating, the maximum heating capacity shall be not less than 4,000 Btu/hr.

3.5.8 Tilted operation. The air conditioner shall start and operate without malfunction when the base of the air conditioner is positioned 5 degress from the horizontal in any plane. During tilted operation, all condensate shall be disposed of through the condensate drain line.

3.6 Refrigeration system. The refrigeration system shall be of the vapor cycle type employing refrigerant conforming to BB-F-1421, type 12 and shall be furnished with a full charge in the system. The moisture content in the system shall not exceed 25 parts per million (ppm) by weight in the liquid phase. The system shall operate between the pressure limits specified in 3.5.3. The entire refrigerant system shall withstand pressures up to 300 psig without any deformation and without leaking more than 0.5 ounce per year.

3.7 Compressor-motor assembly. The compressor-motor assembly shall be of the hermetically-sealed type as defined in ARI Standard 515-60. Operating and physical characteristics of the compressor-motor assembly shall be compatible with all other components of the air conditioner and shall meet all the requirements specified herein. The compressor motor shall be equipped with thermal-overload protection sized to prevent motor operation above the safe operating temperature for which the motor is rated.

3.8 Air-handling components. All air fans shall be statically and dynamically balanced when installed in the air conditioner.

3.9 Fan motors. The evaporator and condenser fan motors shall be equipped with thermal overload protectors sized to prevent motor operation above the safe operating temperature for which the motor is rated.

3.10 Electrical controls. The electrical controls shall provide automatic operation of the air conditioner. Electric controls shall consist of an "OFF-ON" switch, a selector switch, a conditioned-air thermostat, a temperature-overheat thermostat, a shorting plug, a high pressure cut-out switch, and for class 2 air conditioners only, a phase sequence relay.

3.10.1 "OFF-ON" switch. When the "OFF-ON" switch is positioned in the "OFF" setting, the components of the air conditioner shall be inoperative and deenergized. When the "OFF-ON" switch is positioned in the "ON" setting, the components of the air conditioner shall be operative as specified in the setting of the selector switch.

3.10.2 Selector switch. When the "OFF-ON" switch is positioned in the "ON" setting, the selector switch shall operate as follows. When the selector switch is positioned in the "VENT" setting, the evaporator fan motor shall start and operate. When the selector switch is positioned in the "COOL" setting and

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cooling is required, the evaporator fan motor shall continue to operate; the condenser fan motor and the compressor motor shall start and continue to operate; and the electrical heater shall be deenergized. When no cooling is required, the evaporator fan motor shall continue to operate and the condenser fan motor and compressor motor shall not operate and the electrical heaters shall be deenergized. When the selector switch is positioned in the "HEAT" setting, the evaporator fan motor shall continue to operate and the electrical heaters shall be energized. The condenser fan motor and the compressor motor shall not operate when the air conditioner is on the heating cycle.

3.10.3 Conditioned-air thermostat. The conditioned-air thermostat shall control the cooling temperatures automatically by regulating the cycling of the refrigeration system(s) compressor-motor assembly.

3.10.4 Temperature-overheat thermostat. The temperature-overheat thermostat shall control the heating temperatures by deenergizing the air conditioner when the temperature of the heated air reaches 110° F ±15° F.

3.10.5 Shorting plug. The shorting plug shall allow the air conditioner to operate as a single unit without using an auxiliary time delay unit which is required for dual air conditioners' operation.

3.10.6 High pressure cut-out switch. The high pressure cut-out switch shall deenergize the air conditioner when the condensing pressure reaches 325 psig, ±10 psig.

3.10.7 Phase sequence relay. The phase sequence relay on the class 2 air conditioner shall prevent the air conditioner from starting when the power supply to the air conditioner is connected in the out-of-phase condition.

3.11 Auxiliary time delay relay. An auxiliary time delay relay to stagger the starting when two air conditioners are used in parallel shall be furnished in the quantity specified (see 6.2). The auxiliary time delay relay shall have a 30-second ±5-second delay period.

3.12 Wiring. All wiring shall be number coded and marked as shown on the wiring diagrams on the applicable drawings. Markings shall not be damaged or made illegible by oil, water, or handling. The wires shall be secured together in compact harnesses and attached to frame members with insulated clamps at intervals to insure a neat and orderly cable run. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or other components that require removal during maintenance except for actual electrical connections to components. Electrical isolators shall be used between all wiring and metal components to prevent abrasive action to wire insulation.

3.13 Air dampers. The air dampers shall control the return-air and the fresh-air in the evaporator section. The air dampers shall contain damper blades manually adjustable between fully open and fully closed by a control lever and linkage. The air dampers shall be connected so that when the return-air damper is fully open, the fresh-air damper is fully closed; and when

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the return-air damper is fully closed, the fresh-air damper is fully open. The air damper blades shall retain their position settings during operation of the air conditioner.

3.14 Installation accessories. The following installation accessories as shown in either figure 1, 2, or 3, or as otherwise specified (see 6.2) shall be furnished:

a. Interconnecting refrigerant hoses to connect the evaporator section refrigeration system to the condenser section refrigeration system.
b. Electrical connecting cables to electrically connect the evaporator section to the condenser section and, when two air conditioners are interconnected, to connect both condenser sections to the auxiliary time delay relay.

3.15 Vibration. When subjected to the test specified in 4.6.3.14 the air conditioners shall show no evidence of structural damage, misalignment or malfunction of components, leaks, fractures, abnormal vibration, or any other irregular operation.

3.16 Reliability. The air conditioner shall operate for not less than 3,000 hours on the cooling cycle and 100 hours on the heating cycle without failure or malfunctioning and without requiring service or maintenance other than cleaning of the air filter (see 4.6.3.15).

3.17 Storage temperature. The air conditioner shall not be impaired by storage at an ambient temperature of +155° F for 4 hours daily for a period of 2 days and at -65° F without solar radiation for 3 consecutive days.

3.18 Rain, salt fog, and humidity. The air conditioner shall show no evidence of peeling of paint, corrosion, malfunctioning, refrigeration leaks, rough operation, electrical shorts, erratic pressure, or other irregular operation when subjected to rain and salt fog and when exposed to a temperature of 155° F and 95 percent relative humidity.

3.19 Panels. When subjected to rain conditions specified in MIL-STD-810, method 506, all solid panels attached to structural frames shall prevent water from entering the cabinets.

3.20 Electromagnetic interference. The electromagnetic interference emission characteristics of the air conditioner shall confrom to MIL-STD-461A, notice 1, table A-1 of the appendix, when tested as specified in 4.6.3.18 (see 6.6).

3.21 Lubrication. All surfaces requiring lubrication shall be provided with means for lubricating.

3.22 Fungus and moisture resistance. The electrical circuitry including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.22.1), with 1 percent copper-8-quinolinolate or 6 percent salicylanilide (by weight) based on the non-volatile content of the varnish:

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a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose function will be adversely affected by varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.22.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Quality Management District.

3.23 Identification marking. the air conditioner shall be identified in accordance with MIL-STD-130 and as specified in the drawing.

3.24 Instruction plates. The air conditioner shall be equipped with instruction plates including warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the air conditioner. The words "CAUTION" and "WARNING" shall be red, matching color No. 31136 of FED-STD-595. Schematic wiring diagram shall be attached to the air conditioner.

3.25 Stenciling. The gross weight of the air conditioner shall be stenciled on each side of the air conditioner in such a manner as to be discernible. The power for which the air conditioner is wired shall be stenciled on the front of the air conditioner. All stenciling shall conform to MIL-T-704.

3.26 Treatment and painting. Unless otherwise shown on the drawings, all parts shall be cleaned, treated, and painted in accordance with MIL-T-704, type A or type B for aluminum with the following exceptions:

a. Internal components previously painted by the manufacturer need not be repainted if they satisfy the salt-fog requirements.
b. Coils, refrigerant tubing, electrical components, or other components whose operation will be impaired by painting shall not be painted.
c. Dissimilar metals shall be insulated from one another in accordance with MIL-T-704.
d. The color of the finish shall be as specified (see 6.2 and 6.7).

3.27 Class 1, 115-volt, 50/60-hertz, single-phase. The class 1 air conditioner shall be in accordance with Data List DL13211E4726. The weight of the air conditioner when charged with refrigerant and oil shall be not more than 130 pounds.

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3.28 Class 2, 208-volt, 400-hertz, three-phase. The class 2 air conditioner shall be in accordance with Data List DL13211E4600. The weight of the air conditioner when charged with refrigerant and oil shall be not more than 125 pounds.

3.29 Mobility. The air conditioner shall be capable of being transported, without damage, by military van or truck on level tract cross country at variable speeds up to 20 miles per hour when mounted by the mounting holes provided.

3.30 Workmanship.

3.30.1 Metal fabrication. Sheet metal and structural members used in fabrication shall be straight and free from lamination. Sheet metal and structural members may be cut by shearing or sawing. All burrs and rough edges shall be removed. All bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.30.2 Bolted connections. Bolt holes shall be punched or drilled and shall have the burrs removed. Washers and lockwashers shall be provided, and all bolts, nuts, and screws shall be tightened so that they will not work loose under vibration.

3.30.3 Riveted connections. Rivet holes shall be punched or drilled and shall have the burrs removed. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the members.

3.30.4 Welders and welding.

3.30.4.1 Welders. Before assigning any welder to welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder and machine (or method) have passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Welding Qualifications.

3.30.4.2 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, oil, or any other foreign matter. Immediately before welding, oxide shall be removed by an abrading or a chemical process. Grease and oil shall be removed by wiping with a solvent such as acetone. All surfaces shall be dried after cleaning. Welds shall transmit stress without permanent deformation or failure when the parts connected by welds are subjected to proof and service loadings. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part.

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3.30.5 Jigs and fixtures. Shop-fabricated components shall be assembled in steel jigs or frames and joined while held in positon. Jigs and frames shall be designed to minimize distortion.

3.30.6 Interchangeability. All parts having the same part number and like components shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts selection for fit or performance.

3.30.7 Brazing. During the brazing operation, a small amount of non-oxidizing gas shall be continuously bled through the refrigerant lines to prevent internal oxide scaling. Care shall be exercised during the brazing operation to protect all components from deformation and damage. The quantity of brazing alloy and flux used shall be no more than that required to make the joint. Flux shall not be present in the refrigerant system and heavy bends and fillets of braze material shall be avoided.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Air conditioners failing a performance test shall be disassembled by the contractor in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.1.3 Parts and components. Parts and components shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings (see 6.5).

4.1.4 End item final inspection requirements (EIFIR). Each air conditioner shall be inspected in accordance with the applicable EIFIR specified on the drawings. One copy of the completed EIFIR shall be shipped with each inspection air conditioner (see 6.5.2).

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. One first article air conditioner shall be examined as specified in column 2 of table II and the other first article shall be examined as specified in column 3 of table II. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. One first article air conditioner shall be tested as specified in column 2 of table II and the other first article shall be tested as specified in column 3 of table II in the sequence listed. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. An initial production air conditioner(s) will be selected at random by the Government from the first ten air conditioners being produced from production tooling and will be examined and tested as specified in column 5 of table II, to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production air conditioner shall not exclude the remaining air conditioners from the quality conformance inspection and acceptance provisons specified in section 4.

4.4.1 Inspection failure. Failure of an initial production air conditioner to meet any requirement specified herein during and as a result of the examination and tests specifed in 4.4 shall be cause for rejection of the initial production air conditioner(s) and shall be cause for refusal by the Government to continue acceptance of production air conditioners until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiences shall be accomplished by the contractor at no cost to the Government on air conditioners previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all air conditioners accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each air conditioner shall be examined as specified in table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each air conditioner shall be tested as specified in table II in the sequence listed. Failure of any test shall be cause for rejection.

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4.6 Inspection procedure.

4.6.1 Inspection schedule. The inspection schedule is contained in table II.

TABLE II. Inspection schedule.1/

Inspection sequence	Type of inspection				Examination or test	Inspection paragraph	Requirement paragraph
	First article	Preproduction	Quality conformance	Initial production & inspection comparison			
		3	4	5	6	7	8
1	X	X	X	-	Examination.	4.6.2	3.6, 3.12, 3.13, 3.22 thru 3.28, and 3.30
2	X	X	X	-	Refrigeration system leak test.	4.6.3.1	3.6
3	X	X	X	-	Controls test.	4.6.3.2	3.10
4	X	-	X2/	-	Refrigerant moisture content test.	4.6.3.3	3.6
5	X	-	X3/	-	Capacity and airflow tests.	4.6.3.4 4.6.3.4.1	3.5, 3.5.1, 3.5.3 3.5.4, 3.5.5, 3.5.6
6	X	-	-	-	Cycling test.	4.6.3.5	3.5.2
7	X	-	-	-	Heating test.	4.6.3.6	3.5.7
8	X	-	-	-	Evaporator coil frost test.	4.6.3.7	3.5.2
9	-	X	-	-	High temperature test.	4.6.3.8.1	3.5.2, 3.17
10	-	X	-	-	Low temperature test.	4.6.3.9	3.5.2
11	-	X	-	-	Low temperature storage test.	4.6.3.8.2	3.17
12	-	X	-	-	Rain test.	4.6.3.10	3.18
13	-	X	-	-	Humidity test.	4.6.3.12	3.18
14	-	X	-	-	Salt fog test.	4.6.3.11	3.18
15	X	X	-	-	Controls test.	4.6.3.2	3.10
16	X	-	-	X	Motor winding thermal overload protectors tests.	4.6.3.13	3.7, 3.9
17	-	X	-	-	Vibration test.	4.6.3.14	3.15

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TABLE II. Inspection schedule.1/ (Cont'd)

	Type of inspection				Examination or test	Inspection paragraph	Requirement paragraph
1	2	3	4	5	6	7	8
18	X	X	X	-	Refrigeration system leak test.	4.6.3.1	3.6
19	X	X	X3/	-	Controls test.	4.6.3.2	3.10
20	X	-	-	-	Reliability test.	4.6.3.15	3.16
21	X	X	X	-	Refrigeration system leak test.	4.6.3.15.1.1	3.6
22	X	-	-	-	Tilted operation test.	4.6.3.16	3.5.8
23	X	-	X	-	Auxiliary time delay test.	4.6.3.17	3.11
24	X	-	-	-	Electromagnetic interference test.	4.6.3.18	3.20
25	-	-	X	X	Verification examination.	4.6.2	3.6, 3.12, 3.13, 3.22 thru 3.28, and 3.30
26	-	-	X	X	Functional test.	4.6.3.19	3.10
27	-	-	X	X	Operational test.	4.6.3.20	3.10
28	-	-	-	X	Road test.	4.6.3.21	3.29
29	-	-	X	X	Verification examination.	4.6.2	3.6, 3.12, 3.13, 3.22 thru 3.28, and 3.30
30	-	-	-	X	Reliability test.	4.6.3.15	3.16

1/ Perform inspection marked "X".
2/ Test performed on first production unit. Contractor may certify remaining production units provided that the method of evacuating, dehydrating, and charging of refrigerant system is identical to first production unit.
3/ See 4.6.3.15.

4.6.2 Examination. The air conditioner shall be examined for the following defects:

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(First article and quality conformance only)

101. Design dimensions, components, and materials not in accordance with drawings other than those covered by QAP.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as detailed in MIL-T-704 are not effectively insluated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Weight of air conditioner not as specified.
106. Air conditioner not fully charged with the type of refrigerant specified.
107. Electrical circuitry not protected from the effects of moisture and fungus growth.
108. Installation of wiring not as specified.
109. Treatment and painting defects such as blisters, thin spots, lumps, scratches, runs, or sags.
110. Parts or components missing.
111. Technical publications missing or not as specified.
112. Metal fabrication not as specified.
113. Bolted and riveted connections not as specified.
114. Washers or lockwashers missing.
115. Air dampers not operating as specified.
116. Identification marking missing or not as specified.
117. Electromagnetic interference supression of components not identical with approved model.

(Initial production and inspection comparison only)

118. Accessories, power supply plug, manuals, or other components specified in contract or order not furnished.
119. Evidence of exterior or interior damage incurred before or during shipment.
120. Installation mounting nuts missing or not accessible for mounting.
121. Paint flaking, blistering, peeling, or not smoothly applied.
122. Fasteners missing or will not work.
123. Metal shavings or other foreign material inside unit.
124. Air dampers not free and operating as specified.
125. Evidence of refrigerant or oil leaks.
126. Components not mounted securely.
127. Refrigerant tubing not secure; clamps missing.
128. Electrical connections not secure, wires abraded or broken, wires not identified and marked.
129. Gasketing and insulation not secure.
130. Fans binding and not rotating freely.
131. Designation plates, instruction plates and identification plates missing.

4.6.3 Tests. Tests shall be conducted employing the electric power for which the air conditioner will be connected when delivered to the Government. No

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braces, sealing materials, tiedown devices, or other items that are not inherently a part of the air conditioner shall be employed. Unless otherwise specified herein, no alinement of components, addition of refrigerant, or other servicing shall be performed on the first article or production model air conditioners after testing of the air conditioner has been initiated. Prior to test, the air conditioner shall be lubricated and run-in for a period of not less than 1 hour. Components subject to atmoshpheric corrosion shall not be treated with any treatment such as grease or oil in order to meet test requirements. Cooling capacity, airflow, and individual performance tests shall be conducted utilizing the instruments, apparatus, and procedures specified in ASHRAE Standard 16-69 and as specified herein. The tests shall be conducted within a controlled environment so that the desired test conditions can be maintained. The condenser and the evaporator sections of the refrigeration system shall be connected by flexible suction and discharge lines not more than 60 inches in length.

4.6.3.1 Refrigeration system pressure. Charge the refrigeration system to 300 psig, +5 psig and -0 psig, with dry nitrogen containing refrigerant gas detectable with an electronic leak detector, the General Electric, "The Ferret", Model H-25 or approved equal. Calibrate the leak detector for the type of gas and percentage of mixture employed. Check the entire refrigeration system with the above specified leak detector. A leakage rate greater than 0.5 ounce per year or any deformation of the refrigeration system including the condenser and evaporator coils and the compressor case shall constitute failure of this test.

4.6.3.2 Controls. Conduct the controls test at an ambient temperature of between +60° F and 80° F. Ambient temperature shall be determined adjacent to the bulb of the conditioned-air thermostat.

4.6.3.2.1 "OFF-ON" switch. Position the "OFF-ON" switch to the "OFF" position. Determine whether the air conditioner components are deenergized and inoperative. Position the "OFF-ON" switch to the "ON" position. Determine whether the air conditioner components are energized and operative as specified in the setting of the selector switch (see 4.6.3.2.2). Nonconformance to 3.10.1 shall constitute failure of this test.

4.6.3.2.2 Selector switch and conditioned-air thermostat. With the "OFF-ON" switch positioned in the "ON" setting, position the selector switch on the "VENT" setting. Determine whether the evaporator fan motor is operating. Position the selector switch to the "COOL" position and the conditioned-air thermostat at the maximum decrease position. Immerse the bulb of the conditioned-air thermostat in water having a temprature of 80° F. Determine whether the compressor motor, condenser fan and evaporator fan are operating. Determine whether there is a decrease in the temperature of the evaporator air discharge. With the selector switch still on the "COOL" position, position the conditioned-air thermostat at the maximum increase position. Immerse the bulb of the conditioned-air thermostat in water having a temperature of 60° F. Determine whether the evaporator fan motor is operative, the compressor motor and condenser fan motor are inoperative, and that there is no increase in

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temperature of the evaporator air discharge. With the conditioned-air thermostat at the maximum "INCREASE" position, position the selector switch on the "HEAT" position. Immerse the bulb of the conditioned-air thermostat in water having a temperature of 40° F. Determine whether the compressor motor and condenser fan are deenergized and inoperative, whether the evaporator fan motor is operative, and whether the electric heating elements are energized. Repeat the above operations not less than seven times. Nonconformance to 3.10.2 shall constitute failure of this test.

4.6.3.2.3 Temperature-overheat thermostat. Operate the air conditioner on the heating cycle with the selector switch positioned on the "HEAT" setting. After 15 minutes of operation, stop the evaporator fan airflow, thus causing an increase in temperature within the evaporator compartment. Determine whether the electrical heating elements are deenergized. Nonconformance to 3.10.4 shall constitute failure of this test.

4.6.3.2.4 Shorting plug. Install the shorting plug and energize the air conditioner at maximum cooling condition. Inability of the air conditioner to operate shall constitute failure of this test.

4.6.3.2.5 High pressure cut-out switch. Install a pressure gage in the high side of the refrigeration system. With the air conditioner operating on the cooling cycle, decrease the air flow through the condenser coil, thus causing a rise in the compressor-discharge pressure. Inability of the high pressure cut-out switch to deenergize the air conditioner when a condensing pressure of 325 psig, ±10 psig, occurs shall constitute failure of this test.

4.6.3.2.6 Phase-sequence relay. For the class 2 air conditioner, connect the power supply to the air conditioner out of phase. Energize the air conditioner with the "OFF-ON" switch on "ON" position, the selector switch positioned on the "COOL" setting and the conditioned-air thermostat at the maximum increase. Inability of the phase-sequence relay to prevent the air-conditioner from starting shall constitute failure of this test. After completion of this test, restore the power supply to the air conditioner in the proper phase.

4.6.3.2.7 Auxiliary time delay relay. Connect the auxiliary time delay relay between two air conditioners. Turn on both air conditioners simultaneously. Inability of the auxiliary time delay relay to provide a 30-second ±5-second interval of starting between the two air conditioners shall constitute failure of this test.

4.6.3.3 Refrigerant moisture content. After 2 hours of continuous operation of the cooling cycle in a minimum ambient temperature of +95° F, shut down the air conditioner and, after a period of 1 hour, withdraw a sample of refrigerant vapor. Determine the moisture content by one of the following:

a. The methods specified in BB-F-1421.
b. The electric conductometer method employing hygroscopic film.
c. The use of an electrolytic water analyzer.

A moisture content of more than 25 ppm by weight shall constitute failure of this test.

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4.6.3.4 Cooling capacity and airflow. Test the air conditioner for cooling capacity at the capacity rating conditions shown in table I. Test the evaporator fan for airflow at the capacity rating condition specified in 3.5. The air conditioner shall be tested in two sections; the evaporator section shall be located in the evaporator-side room and the condenser section shall be located in the condenser-side room. The sections shall be separated by means of an insulated dividing wall with holes installed to pass through the interconnecting refrigerant hoses and electrical cable and sealed to prevent leakage between the two rooms. In addition to the data required in ASHRAE Standard 16-69, the following data shall be recorded:

a. Compressor suction temperature and pressure.
b. Compressor discharge temperature and pressure.
c. Temperature of refrigerant entering the condenser coil.
d. Temperature of refrigerant leaving condenser coil.
e. Temperature of refrigerant entering thermal expansion valve.
f. Temperature of refrigerant leaving evaporator coil.
g. Certified copy of compressor rating cures for compressor being tested.

Nonconformance to 3.5.1, 3.5.3, 3.5.4, and 3.5.6 shall constitute failure of this test.

4.6.3.4.1 50-hertz operation. When class 1 air conditioners are furnished, the capacity and airflow tests shall be repeated utilizing 115-volt, single-phase, 50-hertz input. Inability of the air conditioner to operate shall constitute failure of this test.

4.6.3.5 Cycling. Cycle the air conditioner on and off on cooling cycle not less than three times at 125° F plus 3° F minus 0° F ambient temperature with air at +125° F plus 3° F minus 0° F entering the evaporator and condenser. Operate the air conditioner for not less than 15 minutes on each cycle. Class 1 air conditioner shall be operated on 50-hertz input. Restart the air conditioner after a waiting period of 3 minutes between cycles. Inability of the air conditioner to restart and operate after each 3-minute waiting period shall constitute failure of this test.

4.6.3.6 Heating. Conduct this test at the prevailing room temperature. Prior to recording any data, operate the air conditioner with the selector switch in the "HEAT" position and the conditioned-air thermostat bulb immersed in ice water for a period of not less than 2 hours. Measure the heating capacity utilizing the following formula:

$$\text{Btu/hr} = \text{cfm}\ 1.08\ (t_2 - T_1)$$

where t_2 = temperature of air leaving heating elements.
t_1 = temperature of air to heating elements.

Nonconformance to 3.5.7 shall constitute failure of this test.

4.6.3.7 Evaporator coil frost. Operate the air conditioner on cooling cycle with the thermostat setting on maximum "DECREASE" position for a continuous

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period of 6 hours in an ambient temperature of +55° F plus or minus 5° F with air at +72° F plus or minus 3° F dry bulb and +60° F plus or minus 3° F wet bulb entering the evaporator. Any evidence of frost or ice forming on the evaporator coil shall constitute failure of this test.

4.6.3.8 Storage temperature.

4.6.3.8.1 High temperature. Subject the air conditioner to a temperature of +155° F for a period of 4 hours. At the conclusion of the exposure period, exmaine the air conditioner for deterioration, corrosion, or damage. After completion of the examination, operate the air conditioner on cooling cycle for a period of not less than 1 hour in an ambient temperature of +125° F plus 3° F minus 0° F with air at +125° F plus 3° F minus 0° F entering the evaporator and the condenser. Turn the air conditioner off for a period of 15 minutes. While maintaining 125° F plus 3° F minus 0° F ambient temperature, restart the air conditioner on cooling cycle with air at +125° F plus 3° F minus 0° F entering the evaporator and the condenser, and operate the air conditioner for a period of not less than 15 minutes. When class 1 air conditioners are furnished, operate the air conditioner at the above conditions utilizing 115-volt, single phase, 50-hertz input. Inability of the air conditioner to start, operate, and restart, or any evidence of deterioration, corrosion, damage, or loss of refrigerant shall constitute failure of this test.

4.6.3.8.2 Low-temperature storage. Subject the air conditioner to a temperature of -65° F plus 0° F minus 3° F for a continuous period of 12 hours. At the conclusion of the exposure period, remove the air conditioner from the test chamber and examine for deterioration and damage. After completion of the examination, allow the air conditioner to soak at room ambient temperature for a period of 1 hour. After completion of the soaking period, operate the air conditioner on cooling cycle for a period of not less than 1 hour. When class 1 air conditioners are furnished, operate the air conditioner at the above conditions utilizing 115-volt, single-phase, 50-hertz input. Inability of the air conditioner to start or operate, or any evidence of deterioration, corrosion, damage, or loss of refrigerant shall constitute failure of this test.

4.6.3.9 Low temperature heating cycle operation test. Operate the air conditioner on heating cycle for a period of not less than 1 hour in an ambient temperature of -65° F plus 0° F minus 3° F. Turn the air conditioner off for a period of 15 minutes. While maintaining the -65° F plus 0° F minus 3° F ambient temperature, restart the air conditioner on heating cycle and operate the air conditioner for a period of not less than 15 minutes. When class 1 air conditioners are furnished, operate the air conditiner at the above conditions utilizing 115-volt, single-phase, 50-hertz input. Inability of the air conditioner to start, operate, or restart shall constitute failure of this test.

4.6.3.10 Rain. Conduct the rain tests in accordance with MIL-STD-810 and as specified herein. The air conditioner evaporator supply and return air openings shall be shielded to prevent the water from entering. Water shall contain fluorescein dye for ease in determining the locations and extent of leakage. After each rain test, remove all removable panels, access doors, and component covers from the air conditioner as required to examine for water penetration.

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4.6.3.10.1 Free rain. With the air conditioner not operating, subject each of the four sides and the top of the air conditioner to a simulated rainfall of 4 inches per hour ±1 inch for 30 minutes each, for a total test duration of not less than 2-1/2 hours. Direction of rainfall against each of the four sides shall be varied throughout the test from vertical to 45 degrees from vertical.

4.6.3.10.2 Basis for failure. Presence of any one of the following conditions shall constitute failure of the rain tests:

a. Malfunctioning of components or an electrical short during operation.
b. Evidence of water penetration into or on any electrical component (such as electrical connector, terminal, switch, or control) that could result in an electrical short, cause damage from rust or corrosion, or become a hazard to personnel.
c. Evidence of water contacting any nonelectrical component (such as thermal insulation) that will absorb or be damaged by water.
d. Evidence of water being trapped inside the air conditioner.

4.6.3.11 Salt fog. Conduct the salt fog test in accordance with MIL-STD-810 and as specified herein. Do not operate the air conditioner during exposure. Following the examination and cleaning at the conclusion of test, operate the air conditioner for not less than 1 hour in an ambient temperature between +60° F and 80° F. Salt deposits resulting from the exposure conditions may be removed by rinsing with tap water prior to operation of the air conditioner. Nonconformance to 3.18 shall constitute failure of this test.

4.6.3.12 Humidity. Conduct the humidity test in accordance with MIL-STD-810 except that the maximum temperature shall be 155° F. Nonconformance to 3.18 shall constitute failure of this test.

4.6.3.13 Thermal overload protectors.

4.6.3.13.1 Compressor motor. Place the air conditioner in an ambient temperature of +25° C plus or minus 1.7° C (77° F plus or minus 3° F) for 24 hours. Measure and record the resistance and temperature of each winding in the motor. Operate the air conditioner on the cooling cycle with the conditioned-air thermostat on maximum increase in an ambient temperature of +125° F plus 3° F minus 0° F with air at +125° F plus 3° F minus 0° F entering the evaporator section. Partially restrict the condenser air discharge opening until the condensing pressure reaches 300 psig or until the compressor motor thermal overload protector is actuated. Determine the temperature of the motor windings using the following formula:

$$T_2 = \frac{259.5R_2 - 234.5R_1}{R_1}$$

where R_1 = resistance of windings at 25° C ambient.
R_2 = resistance of windings at time of actuation of thermal overload device or upon reaching 300 psig.
T_2 = Temperature of motor windings, degrees centigrade.

Motor winding temperature (T_2) in excess of motor manufacturer's recommended limit shall constitute failure of this test.

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4.6.3.13.2 Condenser fan and evaporator fan motors (for class 2 air conditioner only). Place the air conditioner in an ambient temperaure of +77° F for 24 hours. Measure and record the resistance and temperature of each winding in the condenser fan and evaporator fan. Disconnect and isolate the compressor motor. Lock the rotor on the condenser fan and on the evaporator fan with the "OFF-ON" switch on "ON" and the selector switch on "COOL". With the conditioned-air thermostat on maximum, operate both fans until the thermal overloads are actuated. Determine the temperature of the motor windings utilizing the formula specified in 4.6.3.13.1. Inability of the thermal overloads to prevent motor operation above the safe operating temperature for which the motor is rated shall constitute failure of this test.

4.6.3.14 Vibration. Conduct the vibration test as specified in MIL-STD-810, method 514.2, category F, procedure VIII, table 514.2-VI, figure 514.2-6, curve V. The test time shall be 60 minutes total per plane. The vibration test shall be conducted on a vibration machine with both sections of the air conditioner mounted back to back using the side mounting brackets furnished with the air conditioner and by bolts through the four mounting holes located in the base of both sections of the air conditioner. Auxiliary braces, straps, or other tiedown devices shall not be employed to secure the air conditioner to the test apparatus. No blocking or tiedown of internal components other than those that are a part of the air conditioner shall be employed. Removable panels shall be in place during all testing. With the air conditioner mounted, there shall be no resonance inherent in the test apparatus at the frequencies and acceleration level specified for this test. During the tests, the base of the air conditioner shall remain in a horizontal plane. Mount accelerometers on the test apparatus immediately adjacent to the four corners of the base of the air conditioner. The four accelerometers shall be connected into an electric or electronic network which will average the accelerometers output signals into a single indicated g level of the magnitude specified for this test. After completion of each test axis, operate the air conditioner to ensure that all controls and components function. Conduct the refrigeration system leak test specified in 4.6.3.15.1.1. Any evidence of structural damage, misalignment or malfunctioning of components, leaks, abnormal vibration, or any other irregular operation shall constitute failure of this test.

4.6.3.15 Reliability. The values obtained on the first article for this reliability test shall be used as criteria for testing all production air conditioners. In the event the air conditioner does not conform to 4.6.3.15.1, it may be subjected to the test specified in 4.6.3.4.

4.6.3.15.1 Performance. Place the air conditioner within an enclosure maintained at +90° F ±2° F dry bulb. Maintain the air entering the evaporator at 90° F ±2° F dry bulb and at a dry coil condition. Install sheet metal ducts on the evaporator-return air inlet and evaporator-air discharge openings. Each duct shall b a minimum of 12 inches long and of the same cross-sectional dimensions as the respective openings. The return air duct shall have not less than 2 dry bulb thermometers or equal located in the duct 6 inches from the front of the air conditioner and equally spaced over the cross section of the duct at fixed locations so as to give an average entering air temperature. The return air duct shall also have not less than one wet-bulb thermometer located 5

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inches from the front of the air conditioner in the center of the duct, if only one thermometer is used and equally spaced if more than one is used. The evaporator-air discharge duct shall have not less than 4 dry-bulb thermometers located in the duct 6 inches from the front of the air conditioner and equally spaced over the cross section of the duct at fixed locations so as to give an average leaving air temperture; or the evaporator-air discharge duct shall be equipped with an air mixing and sampling device as per ASHRAE standards in order to obtain an average leaving air temperture. Maintain the supply voltage to the air conditioner within ±1 percent during this test. Operate the air conditioner continuously in the cooling mode for not less than 2 hours under the above conditions (class 1 on 60-hertz input). Three sets of the following data shall be recorded and averaged during the last 30 minutes (every 10 minutes).

a. Dry-bulb temperature of air entering the air conditioner, ° F, min. of two thermometers in duct.
b. Dry-bulb temperature of air leaving the air conditioner, ° F, min. of four thermometers in duct or temperature from air sampling device.
c. Wet-bulb temperature of air entering the air conditioner, ° F, min. of one thermometer in duct.
d. Condensing pressure, psig.
e. Suction pressure, psig.
f. Power consumption of the air conditioner, watts.

After completion of the above and while maintaining not less than 90° F dry-bulb temperature entering the evaporator and condenser, operate the air conditioner for a total period of 46 hours in the cooling mode in cyclic operation of 55 minutes "ON" and 5 minutes "OFF". For class 1 air conditioner, 12 hours of the 46 hours shall be on 50-hertz input. After the 46 hours operation, repeat the 2 hour test in the same manner and conditions as specified prior to the 46 hours of operation. Next, operate the air conditioner on the heating cycle (class 1 on 60-hertz input) for 4 hours in an ambient temperature of 75° F ±10° F with air at the same temperature entering the evaporator. During the last hour of operation, record the power input (watts) to the air conditioner.

4.6.3.15.1.1 Leak test. After completion of the reliability test specified in 4.6.3.15, conduct the refrigeration system leak test specified in 4.6.3.1 except that the test pressure shall be the pressure at the prevailing ambient conditions.

4.6.3.15.2 Basis of failure. Any of the following shall constitute failure of the reliability test:

a. Results obtained on each production air conditioner shall not be at variance with the results recorded for the first article (compare results respectively in the order obtained) as follows:

(1) Temperature differential between the average entering dry-bulb temperature and the average leaving dry-bulb temperature more than 1° F less than the first article.
(2) Condensing pressure at variance from first article by more than ±5 psig.

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(3) Suction pressure at variance from first article by more than ±2 psig.
(4) Total watts power consumption on cooling exceeds first article by more than 3 percent.
(5) Total watts power consumption on heating at variance from first article by more than ±5 percent.

b. Failure of any component or fracture or breakdown of any material.
c. Evidence of leakage of refrigerant.
d. Need for any servicing of the air conditioner during operation.

4.6.3.16 Tilted operation. The air conditioner shall be operated 5 degrees from the horizontal position in each of the directions north, south, east, and west. The air conditioner shall operate for a period of 2 hours at rated conditions in each of the four specified positions. Nonconformance to 3.5.8 shall constitute failure of this test.

4.6.3.17 Auxiliary time delay relay. When furnished, connect the auxiliary time delay relay between two air conditioners. Turn on both air conditioners simultaneously. Inability of the auxiliary time delay relay to provide a 30-second ±5-second interval of starting between the two air conditioners shall constitute failure of this test.

4.6.3.18 Electromagnetic interference. The air conditioner shall be tested in accordance with Test Method RE 02 (0.15 MHz to 1000 MHz) and A-CE 1 (1.5 MHz to 65 MHz) as shown in table A-1 of MIL-STD-461A, notice 1, Appendix. The contractor shall furnish the contracting officer a test plan and test report as required by MIL-STD-461. Disapproval of the test report shall constitute failure of this test.

4.6.3.19 Functional. Place the air conditioner in an ambient temperature between 70° F and 80° F. Connect the specified electrical power to the main electrical receptacle of the air conditioner. Position the selector switch on the "OFF" position and determine that the air conditioner components are deenergized and inoperative. Position the selector switch on the "VENT" position and determine that the evaporator fan motor is energized and operating and the condenser fan motor and the compressor are inoperative. Position the selector switch on the "COOL" position and the conditioned-air thermostat at the maximum "DECREASE" position. Determine that the compressor motor and evaporator and condenser fan motors are energized and operating and that there is a decrease in the temperature of the air discharging from the evaporator outlet. Position the conditioned-air thermostat at the maximum "INCREASE" position. Determine that the compressor motor and evaporator and condenser fan motors remain energized and operating. With the conditioned-air thermostat remaining at maximum "INCREASE", position the selector switch on the "HEAT" position. Determine that the evaporator fan motor is energized and operating, that the condenser fan motor and the compressor motor are deenergized and inoperative, and that there is an increase in the temperature of the air discharging from the evaporator outlet. Any one of the following shall constitute failure of this test:

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	Selector Switch Position	Criteria for Failure
a.	"OFF"	Any component energized and operating.
b.	"VENTILATE"	Evaporator fan motor not operating; condenser fan motor and compressor motor operating.
c.	"COOL" with conditioned-air thermostat at maximum "DECREASE" position and at maximum "INCREASE" position.	Compressor motor not operating; evaporator and condenser fan motors not operating and no reduction in temperature of the evaporator air discharge.
d.	"HEAT" with conditioned-air thermostat at maximum "INCREASE" position.	Evaporator fan motor not operating; condenser fan motor and compressor motor operating and no increase in temperature of the evaporator air discharge.

4.6.3.20 Operational. Place the air conditioner in any ambient temperature between 75° F and 125° F. Install sheet metal ducts on the evaporator return-air inlets and evaporator air-discharge openings. Each duct shall be a minimum of 12 inches long and of the same cross-sectional dimensions as the respective openings. Equip each duct with two or more dry-bulb thermometers located in the duct 6 inches from the front of the air conditioner and equally spaced over the cross-section of the duct at fixed locations so as to give an average entering and leaving air temperture. Connect instrumentation to measure voltage and wattage input to the air conditioner. Operate the air conditioner continuously in the maximum cooling mode for not less than two hours. During operation perform the following:

a. Visually observe the refrigerant sight glass to ascertain that the air conditioner has a full charge (clear sight glass).
b. Visually examine for abnoraml operation, such as excessive noise, misalignment of components, or any other irregular operation.
c. Record air temperatures, evaporator inlet and outlet, dry bulb °F.
d. Record volts and amps input to air conditioner.

Then operate the air conditioner on the heating cycle for thirty minutes in an ambient temperature of not more than 80° F and record the voltage and watts input. Any one of the following shall constitute failure of this test:

a. Failure to operate. (Cooling and heating cycle.)
b. Abnormal operation. (Cooling and heating cycle.)
c. Bubbles in sight glass or sight glass not clear. (Cooling cycle.)
d. Air temperature drop less than 20° F between evaporator inlet and outlet. (Cooling cycle.)
e. Current input on cooling cycle and heating cycle more than the following:

Class 1 - 15 Amps
Class 2 - 12 Amps

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4.6.3.21 Road test. Conduct the road testing using a military van or truck. Mount the air conditioner back to back on the van or truck using the four mounting holes located on the bottom of the air conditioner and the flange mounting hole located on each side of the air conditioner. Do not use any other means of mounting the air conditioner, such as auxiliary braces or the tiedown devices. Subject the air conditioner to 300 miles of level track cross-country travel at variable speeds up to 20 miles per hour. After completion of the road test, conduct the verification inspection, functional test and operational test. Failure of any test shall constitute failure of the road test.

4.7 Inspection comparison. The Government may select air conditioners at any time during the contract production period and subject these air conditioners to the examination and tests specified in 4.4 to determine that the quality of the selected air conditioners is equal to the quality standards established during initial production testing. The inspection will be performed by the Government at a site selected by the Government. Air conditioners will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article and initial production air conditioners.

4.7.1 Inspection failure. Failure of any air conditioner when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of air conditioners until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of packaging.

4.8.1 Examination of first article pack. The first article pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For purposes of inspection a completed pack prepared for shipment shall be the unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

132.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.	
133.	Condensor and evaporator sections not secured to bottom of box as specified for level A.	

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134. Technical publications not secured to the condenser section as specified for level A or B.
135. Components not placed and secured in the packing box as specified for level A or B.
136. Blocking, bracing and cushioning not in accordance with the referenced document specified for level A or B.
137. Top of box not secured with screws as specified for level A.
138. Strapping not as specified for level A or B.
139. Marking not as specified for level A, level B, or commercial.
140. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in (see 4.8) and shall be subject to surveillance and approval by the Government (see 6.3). The first article pack may be accomplished utilizing either the first article air conditioner or the production air conditioner. If the first article is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article and the production model.

5.2 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Evaporator and condenser sections. Each section of the air conditioner shall be preserved in accordance with MIL-P-116, method IIa. Each section shall be secured to the bottom of the box specified in 5.3.1 by bolting directly through the barrier material.

5.2.1.2 Technical publications. Technical publications shall be inclosed in a protective envelope and secured to the condenser section prior to enclosing the section in the method II barrier.

5.2.1.3 Auxiliary time delay relay. The auxiliary time delay relay, when included, shall be preserved in accordance with MIL-P-116, method IIb.

5.2.1.4 Cable and hose. Cables and hoses shall be coiled, tied with cotton tape, and preserved together in accordance with MIL-P-116, method III. Power connector and coupling halves 13211E4691 shall be packed and preserved in accordance with MIL-P-116. The box shall conform to PPP-B-636, type CF, class weather-resistant and shall be of a size that will fit between the sections of the air conditioner when they are mounted as specified in 5.3.

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5.2.2 Level B. The evaporator and condenser sections shall not require preservation. Technical publications shall be placed in a protective envelope and secured to the condenser section. The auxiliary time delay relay shall be preserved in accordance with MIL-P-116, method III. Cable and hose shall be preserved in accordance with MIL-P-116, method III in a box which will fit between the sections of the air conditioner when they are packed as specified in 5.3. The box shall conform to PPP-B-636, type CF, class domestic, grade 175 and shall be sealed with tape in accordance with the box specification.

5.2.3 Commercial. Each complete air conditioner shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.3.1 Level A. The air conditioner and components, preserved as in 5.2, shall be packed in a box conforming to PPP-B-621, class 2, style 2, 2-1/2, 3, or 7; or in a box conforming to PPP-B-601, overseas type, style optional. The box shall be modified with the addition of skids in accordance with the box specification. The condenser and evaporator sections shall be secured with bolts to the bottom of the box with sufficient separation to permit the preserved hoses and cables to be placed between the sections. The auxiliary time delay relay shall be placed on top of the evaporator section and secured in a manner to prevent damage to the water-vaporproof barrier. Bundled angles shall be secured in one corner of the box. Tie-down, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. The lid of the box shall be secured with screws instead of nails. Strapping shall conform to QQ-S-781, class I, type I or IV, finish A.

5.3.2 Level B. The air conditioner and components, preserved as in 5.2, shall be packed in a close fitting box conforming to either PPP-B-640, class 2, style E, or PPP-B-636, V11c, V13c or V15c, style optional. The box shall be large enough to permit the preserved hoses and cables to be placed between the two sections of the air conditioner. Appropriate blocking, bracing, and cushioning in accordance with MIL-STD-1186, shall be used to fill voids between the air conditioner and the interior surfaces of the box. Closure of the box shall be in accordance with the appendix to the box specification using steel strapping conforming to QQ-S-781, class I, type I or IV, 5/8-inch wide and 0.020-inch thick.

5.3.3 Commercial. The air conditioner and components, preserved as in 5.2, shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection (level A or level B) shall be in accordance with MIL-STD-129.

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5.4.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

5.4.3 Special marking. For all degrees of packing, special marking shall be placed on shipping containers to indicate that the air conditioner shall remain in an upright position during shipment and storage. Special marking shall consist of arrows and the words "THIS SIDE UP" which shall be placed on two sides of the container in black letters not less than one inch high.

6. NOTES.

6.1 Intended use. The air conditioner is intended for use in providing cooled or heated air to maintain adequate working temperatures in military vans. The air conditioner may be installed indoors or may be exposed to the weather.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. When number of first article is other than specified (see 3.2).
e. Whether initial production air conditioner is required (see 3.3).
f. Number of air conditioners to be furnished for initial production testing when applicable (see 3.3).
g. Quantity of auxiliary time delay relays, if required (see 3.11).
h. Whether installation accessories shall be as shown on figure 1, 2, or 3, or whether they shall be otherwise (see 3.14).
i. Color of finish coat of paint (see 3.26).
j. Time frame required for submission of the first article pack (see 5.1).
k. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the air conditioners in accordance with this specification.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

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6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.5.2 End item final inspection requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

6.6 Electromagnetic interference.

a. These air conditioners have been previously qualified to MIL-STD-461A, Notice 1, and MIL-STD-462, Notice 1, as specified in sections 3 and 4 of this specification. Subsequant notices or revisions should not be made applicable to a procurement contract.
b. The contracting officer should include the following in procurement documents:

(1) Notices or revisions other than those listed below are not applicable:

MIL-STD-461A, Notice 1.
MIL-STD-462, Notice 1.

(2) Bidders are cautioned against failing to properly take into consideration the radio frequency interference suppression/ electromagnetic interference characteristics requirements for the end items in preparing their bids.

The foregoing are performance requirements only. Specific drawings which may be incorporated therein are intended as an example of a method which was successful in the past. Since each offeror has latitude in selecting electrical components within the requirements of the drawings and specifications, and the electromagnetic emission and suppression characteristics may differ from those previously used, the Government does not expressly or impliedly warrant that performance requirements will be met if such drawings are followed and reliance upon such drawings is at the contractor's own risk.

6.7 Finish paint coat validation. Prior to acceptance of any finish coat, the contracting officer should require validation of spectral reflectance characteristics of the finish coat from the US Army Belvoir Research and Development Center, ATTN: STRBE-VO, Fort Belvoir, VA 22060.

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6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each air conditioner.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4120-A288

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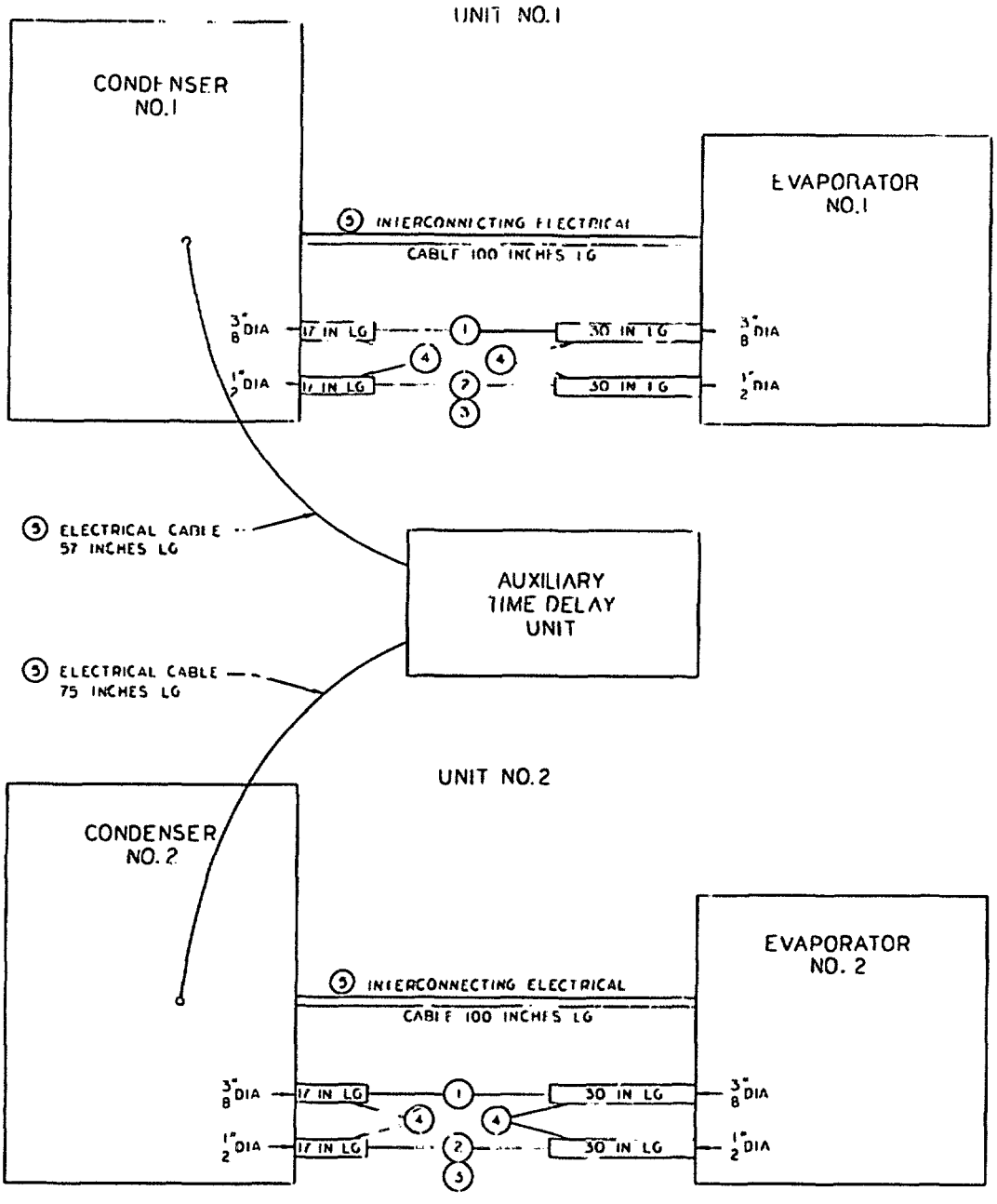


FIGURE 1. INSTALLATION ACCESSORIES FOR 6,000 BTU/HR AIR CONDITIONER PHRSHING SYSTEM, AN/TRC 60, 2 UNITS PER SHELTER

(1) COPPER TUBING 3/8" OD x .180 IN LG FURNISHED BY USER

(2) COPPER TUBING 1/2" OD x .180 IN LG FURNISHED BY USER

(3) FITTINGS TO ATTACH COPPER TUBE TO REFRIGERANT HOSES FURNISHED BY USER

(4)(5) FLEXIBLE REFRIGERANT HOSES AND ELECTRICAL CABLES TO BE FURNISHED WITH AIR CONDITIONER

X-294

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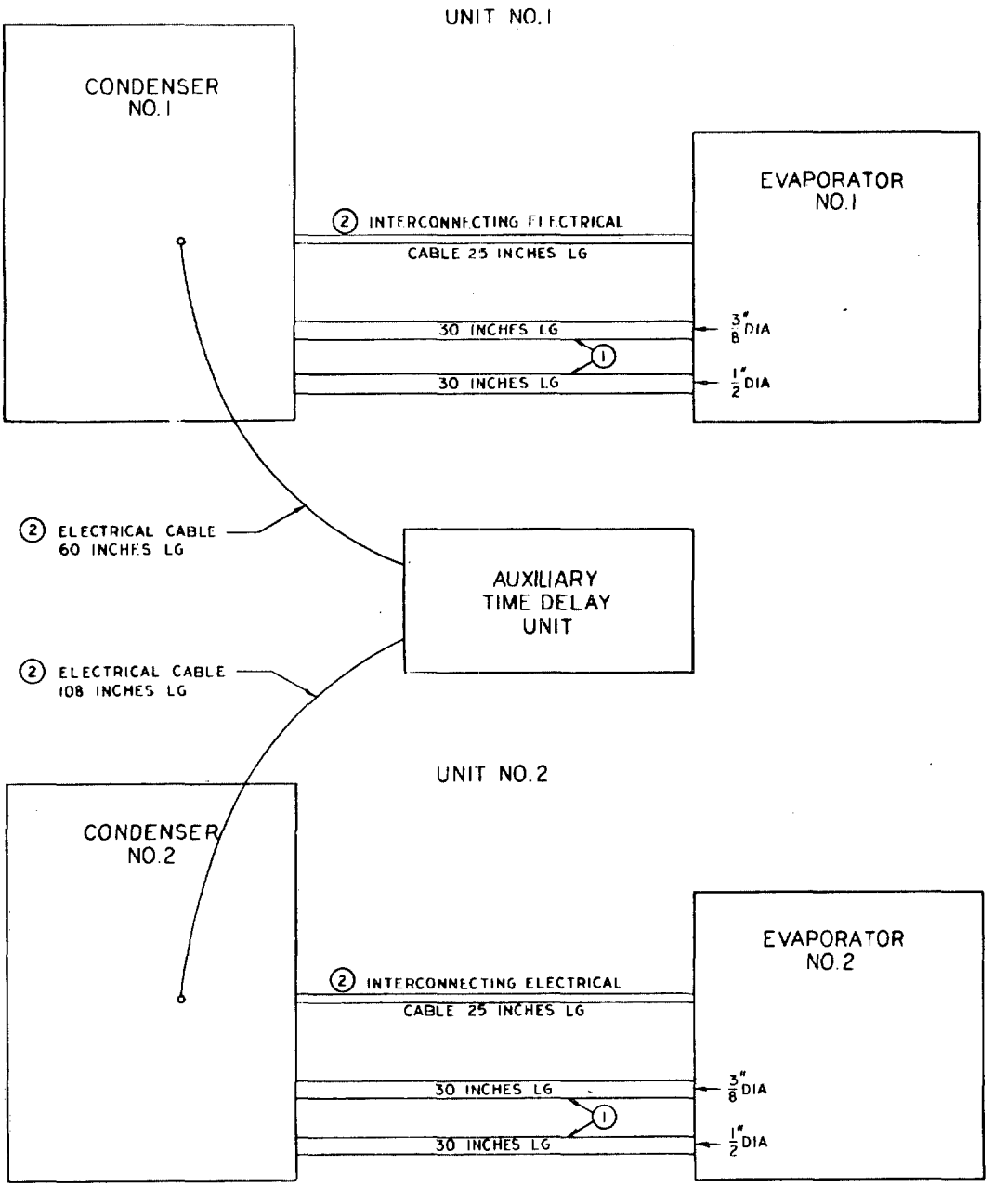


FIGURE 2. INSTALLATION ACCESSORIES FOR 6,000 BTU/HR AIR CONDITIONER-PERSHING SYSTEM, PROGRAMMER TEST STATION, 2 UNITS PER SHELTER

(1)(2) FLEXIBLE REFRIGERANT HOSES AND ELECTRICAL CABLES TO BE FURNISHED WITH AIR CONDITIONER

M-295

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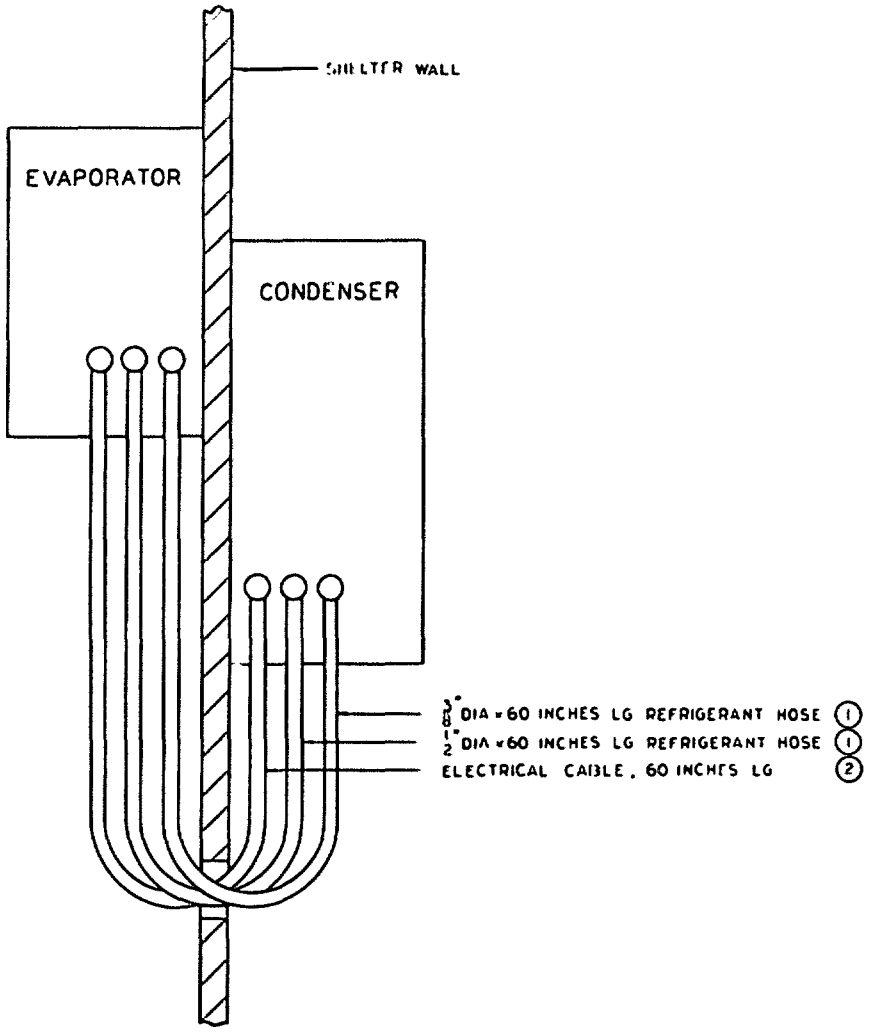


FIGURE 3. INSTALLATION ACCESSORIES FOR AIR CONDITIONER-AN/UAS 4 SYSTEM, AN/TAQ 1, 1 UNIT PER SHELTER

(1)(2) FLEXIBLE REFRIGERANT HOSES AND ELECTRICAL CABLES TO BE FURNISHED WITH AIR CONDITIONER

X-296

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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